

Software Requirements Specification (SRS)

Student Attendance Management System

Field	Details
Project	Student Attendance Management System
Version	1.0
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1.0	03-09-2026	<Team>	Initial SRS prepared for Student Attendance Management System	<Pending>

Approvals

Role	Name	Signature / Email	Date
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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Student Attendance Management System (SAMS). It defines the functional requirements, non-functional requirements, interfaces, security objectives, acceptance criteria, system models, and traceability information required to design, develop, test, and evaluate the system.

1.2 Scope

The Student Attendance Management System is a software application for recording, maintaining, monitoring, and reporting student attendance. The system supports administrators, faculty members, and students. Faculty can mark and update attendance for assigned classes; administrators can manage students, faculty, subjects, classes, and attendance rules; students can view their attendance history and percentage. The system can generate attendance reports and identify students whose attendance falls below a configured threshold. The project excludes payroll, fee collection, examination evaluation, and other unrelated academic functions.

1.3 Audience

The intended audience includes student developers, faculty users, administrators, QA/test engineers, project guides, course coordinators, and assessment evaluators.

1.4 Definitions

Term	Definition
SAMS	Student Attendance Management System
Admin	Authorized user who manages master data, users, classes, subjects, and configuration.
Faculty	Authorized teaching staff member who records and reviews attendance.
Student	Learner whose attendance is recorded and reported by the system.
Attendance Session	A particular class/lecture occurrence for which attendance is recorded.
Attendance Percentage	Percentage of attended sessions out of total conducted sessions, based on the configured rule.
FR	Functional Requirement
NFR	Non-Functional Requirement
RTM	Requirements Traceability Matrix
RBAC	Role-Based Access Control

2. Overall Description

2.1 Product Perspective

SAMS is a centralized web-based application backed by a relational database. It provides role-based access through a web interface and stores student, faculty, subject, class, attendance, and report data. The system may be deployed on a college server or cloud environment and can be integrated with an existing student information system in a future version.

2.2 Major Product Functions (Detailed)

- Secure login and role-based authentication for Admin, Faculty, and Student users.
- Student registration, update, search, activation/deactivation, and record management.
- Faculty profile and account management.
- Subject, class/section, academic semester, and faculty-subject assignment management.
- Attendance session creation and attendance marking by authorized faculty.

- Attendance update/correction with authorization and audit information.
- Automatic calculation of attendance percentages.
- Student-wise, subject-wise, class-wise, and date-range attendance reports.
- Low-attendance identification based on a configurable threshold.
- Dashboard views for attendance summaries and pending actions.
- Search and filtering of attendance records.
- Export of authorized attendance reports in a common report format.
- Audit logging of important administrative and attendance changes.

2.3 User Roles and Characteristics (Expanded)

Role	Characteristics / Responsibilities
Administrator	Manages users, students, faculty, subjects, classes, academic terms, attendance rules, and reports. Requires administrative knowledge.
Faculty	Marks and reviews attendance for assigned classes/subjects, makes authorized corrections, and views attendance reports for their classes.
Student	Views personal attendance percentage, subject-wise summary, attendance history, and low-attendance status. Cannot modify attendance.

2.4 Operating Environment

Client: modern web browser on desktop, laptop, tablet, or mobile device. Server: application server capable of running the selected backend framework. Database: relational database such as MySQL/PostgreSQL. Network: institutional LAN/Wi-Fi or secure Internet connection.

2.5 Constraints

- Only authorized users shall access protected functions.
- Attendance changes shall be restricted according to role and configured policy.
- The database must maintain referential integrity among students, faculty, subjects, classes, and attendance records.
- The system shall operate within the hardware, network, and software resources available to the project.
- Privacy of student information shall be maintained according to applicable institutional policies.
- The project shall use the tools, frameworks, and database approved by the course/project guide.

3. External Interface Requirements

3.1 User Interfaces

The primary interface shall be a responsive web UI with a login page, role-specific dashboard, navigation menu, data entry forms, attendance marking screen, searchable tables, report screens, validation messages, and confirmation prompts. The interface shall display clear success/error messages and prevent accidental duplicate submission where practical.

3.2 Hardware Interfaces

No specialized hardware is mandatory for the baseline project. The system shall operate on standard computers and mobile devices with keyboard/touch input, display, and network connectivity. Optional biometric or QR hardware is outside the baseline scope unless approved as an extension.

3.3 Software Interfaces

- Relational database management system for persistent data storage.
- Backend/application framework for business logic and authentication.

- Web browser for client-side access.
- Optional email/notification service for future attendance alerts.
- Optional CSV/PDF reporting library for authorized report export.

3.4 Communications

Client-server communication shall use HTTPS in production. Requests shall use standard web protocols and authenticated sessions. Database connections shall be protected through server-side configuration and shall not expose database credentials to clients.

4. System Features (Detailed)

Each requirement below is written as a verifiable 'shall' statement and includes acceptance criteria and a test case reference. Requirement IDs follow SAM-F-###.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SAM-F-001	The system shall authenticate users using valid credentials and establish a session associated with the user's role.	Functional	High	Security / Admin	Valid credentials allow login; invalid credentials are rejected. Test: TC-AUTH-01	Requires user account store and authentication module.
SAM-F-002	The system shall provide role-based access for Admin, Faculty, and Student users.	Functional	High	Admin / Security	Each role sees only authorized menus and functions. Test: TC-AUTH-02	Depends on RBAC configuration.
SAM-F-003	The system shall allow administrators to add, update, search, activate, and deactivate student records.	Functional	High	Admin	CRUD operations work and validation prevents incomplete/duplicate records. Test: TC-STU-01	Student database table required.
SAM-F-004	The system shall allow administrators to create and maintain faculty accounts and profiles.	Functional	High	Admin	Faculty can be created, updated, activated/deactivated, and searched. Test: TC-FAC-01	Requires role and user management.
SAM-F-005	The system shall allow administrators to create and maintain academic terms, subjects, classes/sections, and their relationships.	Functional	High	Admin	A subject/class can be created and linked to an academic term. Test: TC-MASTER-01	Referential integrity required.
SAM-F-006	The system shall allow administrators to assign faculty members to authorized subjects and classes.	Functional	High	Admin / Faculty	Assigned faculty can access the mapped class; unassigned faculty cannot. Test: TC-ASSIGN-01	Depends on faculty-subject-class mapping.
SAM-F-007	The system shall allow authorized faculty to create	Functional	High	Faculty	A valid session is created with required	Requires active class, subject, and faculty assignment.

	an attendance session for a selected class, subject, date, and period.				metadata. Test: TC-ATT-01	
SAM-F-008	The system shall display the enrolled students of the selected class and allow faculty to mark each student Present or Absent.	Functional	High	Faculty	Attendance can be recorded for every enrolled student and saved successfully. Test: TC-ATT-02	Requires enrollment/student records.
SAM-F-009	The system shall prevent duplicate attendance records for the same student and attendance session.	Functional	High	Data Integrity	A second attendance submission for the same session/student is rejected or treated as an update. Test: TC-ATT-03	Unique database constraint/business rule.
SAM-F-010	The system shall allow authorized faculty to edit attendance records within the configured correction period and shall record the change.	Functional	High	Faculty / Admin	Authorized edits succeed; unauthorized/out-of-period edits are blocked. Test: TC-ATT-04	Depends on policy and audit logging.
SAM-F-011	The system shall calculate attendance percentage automatically for each student overall and for each subject.	Functional	High	Faculty / Student	Percentage matches the configured formula using stored attendance sessions. Test: TC-CALC-01	Requires accurate attendance data.
SAM-F-012	The system shall allow students to view their own attendance summary and detailed attendance history.	Functional	High	Student	Student sees only their own records with correct dates, subjects, and status. Test: TC-STU-02	Requires authenticated student session.
SAM-F-013	The system shall allow faculty to view attendance summaries and detailed records for their assigned classes and subjects.	Functional	High	Faculty	Faculty can filter and view only assigned classes/subjects. Test: TC-FAC-02	Depends on authorization mapping.
SAM-F-014	The system shall allow administrators to generate student-wise, subject-wise, class-wise, and date-range attendance reports.	Functional	Medium	Admin / Faculty	Reports reflect selected filters and stored data. Test: TC-REP-01	Reporting module required.
SAM-F-015	The system shall identify students whose attendance	Functional	High	Admin / Faculty	Students below threshold are correctly flagged in	Requires configurable threshold.

	percentage is below the configured threshold.				dashboard/report. Test: TC-ALERT-01	
SAM-F-016	The system shall allow authorized users to search and filter attendance records by student, subject, class, date, and attendance status.	Functional	Medium	Admin / Faculty	Filters return only matching records. Test: TC-SEARCH-01	Depends on indexed/searchable fields.
SAM-F-017	The system shall allow authorized users to export attendance reports in a configured downloadable format such as CSV or PDF.	Functional	Medium	Admin / Faculty	Export contains the same authorized records shown by the report filters. Test: TC-EXP-01	Requires reporting/export library.
SAM-F-018	The system shall maintain an audit record for important actions including attendance creation, correction, user changes, and configuration changes.	Functional	High	Security / Admin	Audit record contains actor, timestamp, action, and affected record identifier. Test: TC-AUDIT-01	Audit logging must be enabled and protected.

5. Non-functional Requirements (Detailed)

NFRs are measurable and tied to test plans. IDs follow SAM-NF-####.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
SAM-NF-001	The system shall return normal user-interface requests within 3 seconds for at least 90% of requests under the defined normal-load test.	Performance	High	90th percentile response time <= 3s. Test: TC-PERF-01
SAM-NF-002	The system shall achieve at least 99% monthly availability excluding scheduled maintenance for the project deployment environment.	Reliability	High	Monthly uptime >= 99%. Test: TC-REL-01
SAM-NF-003	The system shall protect stored credentials using a strong one-way password hashing mechanism and shall not store passwords in plaintext.	Security	High	Database/security review confirms no plaintext passwords. Test: TC-SEC-01
SAM-NF-004	The system shall maintain data integrity so that attendance records cannot	Data Integrity	High	Integrity tests show no orphan attendance records. Test: TC-DATA-01

	reference nonexistent students, sessions, or classes.			
SAM-NF-005	The system shall provide a responsive and understandable interface usable on common desktop and mobile screen sizes.	Usability	Medium	UI review on at least one desktop and one mobile viewport passes. Test: TC-UX-01
SAM-NF-006	The system shall support maintainable modular code with documented database schema, APIs/components, and configuration.	Maintainability	Medium	Code/documentation review checklist passes. Test: TC-MAINT-01
SAM-NF-007	The system shall support at least 100 concurrent authenticated users in the defined load-test environment without critical failure.	Scalability	Medium	Load test completes with no critical errors and target response time is maintained. Test: TC-LOAD-01

5.1 Security

5.1.1 Security Objectives

- Confidentiality: Ensure that student personal information, credentials, and attendance records are accessible only to authorized users.
- Integrity: Prevent unauthorized or accidental modification of attendance and academic master data and maintain a trustworthy audit trail.
- Availability and Accountability: Keep attendance information available to legitimate users and provide sufficient audit information to trace important actions.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
SAM-SR-001	The system shall enforce HTTPS/TLS for all production client-server communications.	Security	High	Network scan/configuration review confirms insecure HTTP access is redirected or disabled. Test: TC-SEC-02
SAM-SR-002	The system shall store user passwords only as salted, strong one-way hashes and shall never log plaintext passwords.	Security	High	Security/code/database review finds no plaintext password storage or logging. Test: TC-SEC-03
SAM-SR-003	The system shall enforce role-based authorization on every protected operation and shall deny unauthorized access by default.	Security	High	Attempted cross-role access is rejected. Test: TC-SEC-04
SAM-SR-004	The system shall validate and sanitize user inputs and use parameterized database operations to reduce injection and malformed-input risks.	Security	High	Security test inputs do not alter queries or bypass validation. Test: TC-SEC-05

SAM-SR-005	The system shall create protected audit logs for attendance changes, account changes, and administrative configuration changes, including actor and timestamp.	Security	High	Audit events are created and cannot be modified by ordinary users. Test: TC-SEC-06
SAM-SR-006	The system shall terminate or expire inactive authenticated sessions according to a configurable timeout policy.	Security	Medium	Inactive session becomes invalid after configured timeout. Test: TC-SEC-07

6. Quality Attributes & Acceptance Tests

The system shall be accepted when all high-priority functional requirements are implemented and verified, security requirements pass, no critical defects remain open, core data integrity checks pass, and the RTM shows coverage for all requirements.

Quality Attribute	Acceptance Test / Exit Criterion
Functionality	All high-priority FRs pass their referenced test cases.
Performance	TC-PERF-01 and TC-LOAD-01 meet stated thresholds in the test environment.
Reliability	System remains stable during the defined operational/load test.
Security	TC-SEC-01 through TC-SEC-07 pass; no critical security defect remains.
Usability	Representative Admin, Faculty, and Student workflows can be completed without blocking UI defects.
Data Integrity	No orphan attendance records; attendance calculations match independently calculated expected values.
Traceability	Every FR, NFR, and security requirement maps to a design/module and at least one test case.

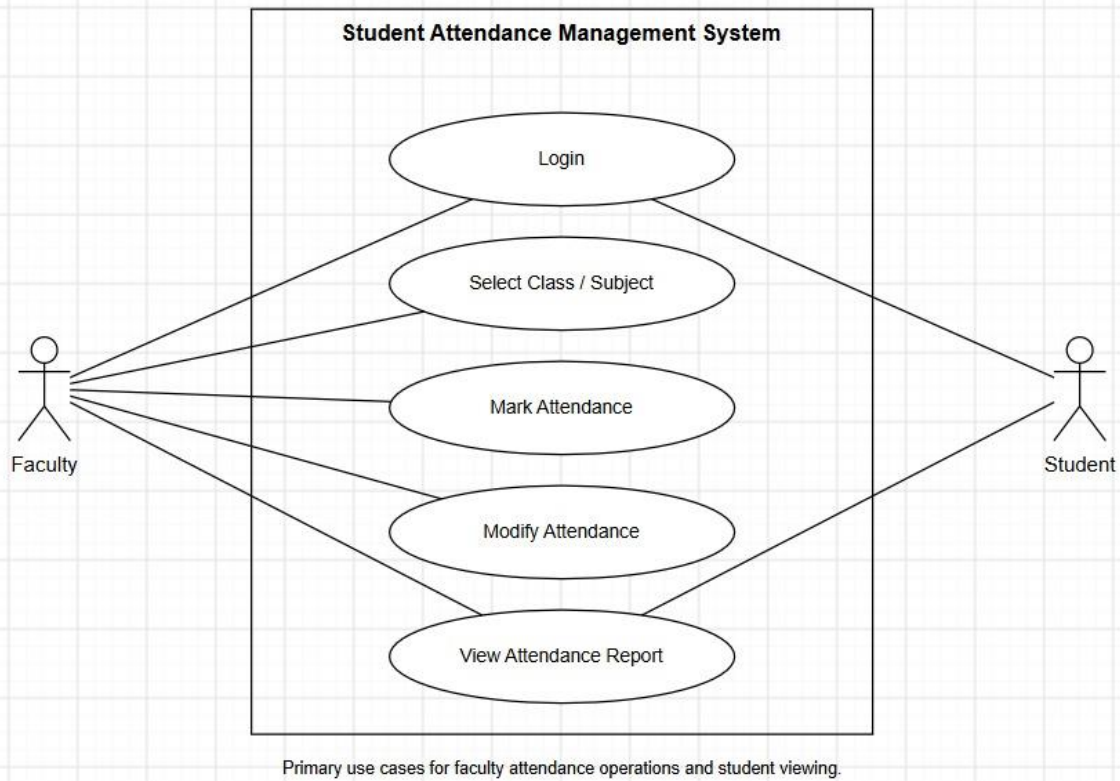
Acceptance Test Suites

- Authentication and role authorization tests.
- Student/faculty/master-data management tests.
- Attendance session creation and marking tests.
- Attendance correction and duplicate-prevention tests.
- Attendance calculation and low-attendance identification tests.
- Reporting, filtering, and export tests.
- Performance, load, reliability, usability, and security tests.

7. System Models and Diagrams

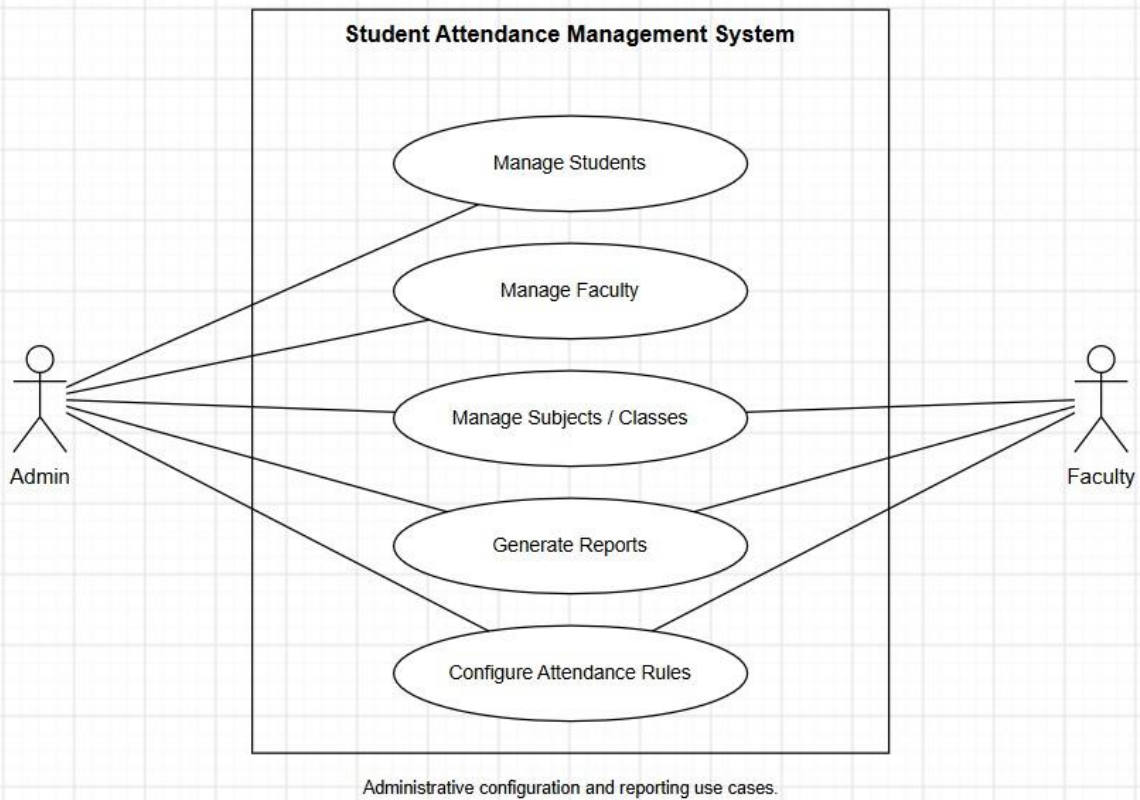
7.1 UML Use-Case Diagram 1 – Attendance Operations

This diagram shows the main interactions of Faculty and Student users with the attendance-related functions.



7.2 UML Use-Case Diagram 2 – Administration and Reporting

This diagram shows administrative management and reporting functions, including master-data configuration.



8. Requirements Traceability Matrix (RTM)

Status values: N = Not implemented/not tested, P = Partially implemented/tested, A = Accepted. The initial SRS status is N.

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
SAM-F-001	The system shall authenticate users using valid	4 / DS-AuthModule	AuthModule	TC-AUTH-01	N	
SAM-F-002	The system shall provide role-based access for A	4 / DS-AuthModule	AuthModule	TC-AUTH-02	N	
SAM-F-003	The system shall allow administrators to add, up	4 / DS-StudentModule	StudentModule	TC-STU-01	N	
SAM-F-004	The system shall allow administrators to create	4 / DS-FacultyModule	FacultyModule	TC-FAC-01	N	
SAM-F-005	The system shall allow administrators to create	4 / DS-MasterDataModule	MasterDataModule	TC-MASTER-01	N	
SAM-F-006	The system shall allow	4 / DS-AssignmentModule	AssignmentModule	TC-ASSIGN-01	N	

	administrators to assign					
SAM-F-007	The system shall allow authorized faculty to cre	4 / DS-AttendanceModule	AttendanceModule	TC-ATT-01	N	
SAM-F-008	The system shall display the enrolled students o	4 / DS-AttendanceModule	AttendanceModule	TC-ATT-02	N	
SAM-F-009	The system shall prevent duplicate attendance re	4 / DS-AttendanceModule	AttendanceModule	TC-ATT-03	N	
SAM-F-010	The system shall allow authorized faculty to edi	4 / DS-AttendanceModule	AttendanceModule	TC-ATT-04	N	
SAM-F-011	The system shall calculate attendance percentage	4 / DS-CalculationModule	CalculationModule	TC-CALC-01	N	
SAM-F-012	The system shall allow students to view their ow	4 / DS-StudentPortal	StudentPortal	TC-STU-02	N	
SAM-F-013	The system shall allow faculty to view attendanc	4 / DS-FacultyPortal	FacultyPortal	TC-FAC-02	N	
SAM-F-014	The system shall allow administrators to generat	4 / DS-ReportModule	ReportModule	TC-REP-01	N	
SAM-F-015	The system shall identify students whose attenda	4 / DS-AlertModule	AlertModule	TC-ALERT-01	N	
SAM-F-016	The system shall allow authorized users to searc	4 / DS-ReportModule	ReportModule	TC-SEARCH-01	N	
SAM-F-017	The system shall allow authorized users to expor	4 / DS-ReportModule	ReportModule	TC-EXP-01	N	
SAM-F-018	The system shall maintain an audit record for im	4 / DS-AuditModule	AuditModule	TC-AUDIT-01	N	
SAM-NF-001	The system shall return normal user-interface re	5 / DS-NFR-001	PerformanceModule	Test: TC-PERF-01	N	
SAM-NF-002	The system shall achieve at least 99% monthly av	5 / DS-NFR-002	ReliabilityModule	Test: TC-REL-01	N	
SAM-NF-003	The system shall protect stored credentials usin	5 / DS-NFR-003	SecurityModule	Test: TC-SEC-01	N	
SAM-NF-004	The system shall maintain data integrity so that	5 / DS-NFR-004	Data IntegrityModule	Test: TC-DATA-01	N	
SAM-NF-005	The system shall provide a responsive and unders	5 / DS-NFR-005	UsabilityModule	Test: TC-UX-01	N	

SAM-NF-006	The system shall support maintainable modular co	5 / DS-NFR-006	MaintainabilityModule	Test: TC-MAINT-01	N	
SAM-NF-007	The system shall support at least 100 concurrent	5 / DS-NFR-007	ScalabilityModule	Test: TC-LOAD-01	N	
SAM-SR-001	The system shall enforce HTTPS/TLS for all produ	5.1 / DS-SEC-001	SecurityModule	Test: TC-SEC-02	N	
SAM-SR-002	The system shall store user passwords only as sa	5.1 / DS-SEC-002	SecurityModule	Test: TC-SEC-03	N	
SAM-SR-003	The system shall enforce role-based authorizatio	5.1 / DS-SEC-003	SecurityModule	Test: TC-SEC-04	N	
SAM-SR-004	The system shall validate and sanitize user input	5.1 / DS-SEC-004	SecurityModule	Test: TC-SEC-05	N	
SAM-SR-005	The system shall create protected audit logs for	5.1 / DS-SEC-005	SecurityModule	Test: TC-SEC-06	N	
SAM-SR-006	The system shall terminate or expire inactive au	5.1 / DS-SEC-006	SecurityModule	Test: TC-SEC-07	N	

Appendix A – Suggested Initial Database Entities

Entity	Purpose	Key fields (illustrative)
User	Authentication and account/role information	UserID, Username, PasswordHash, Role, Status
Student	Student master record	StudentID, Name, Email, ClassID, Status
Faculty	Faculty master record	FacultyID, Name, Email, Status
Subject	Subject master record	SubjectID, SubjectCode, SubjectName
Class/Section	Class or section information	ClassID, ClassName, AcademicTerm
FacultyAssignment	Maps faculty to subjects/classes	AssignmentID, FacultyID, SubjectID, ClassID
AttendanceSession	One class occurrence	SessionID, AssignmentID, Date, Period
AttendanceRecord	Student attendance for a session	AttendanceID, SessionID, StudentID, Status, UpdatedAt
AttendanceRule	Configured attendance threshold/policy	RuleID, ThresholdPercent, CorrectionWindow
AuditLog	Security/accountability history	AuditID, UserID, Action, Entity, EntityID, Timestamp

Appendix B – Assumptions

- Students are enrolled in classes before attendance is marked.
- Faculty can mark attendance only for assigned subjects/classes.
- The attendance percentage formula is configurable; the baseline calculation is attended sessions divided by conducted sessions multiplied by 100.
- The institution will define the official low-attendance threshold and correction policy.
- Internet/network access is available for normal system operation.

- The project will use test/demo data during development and evaluation.